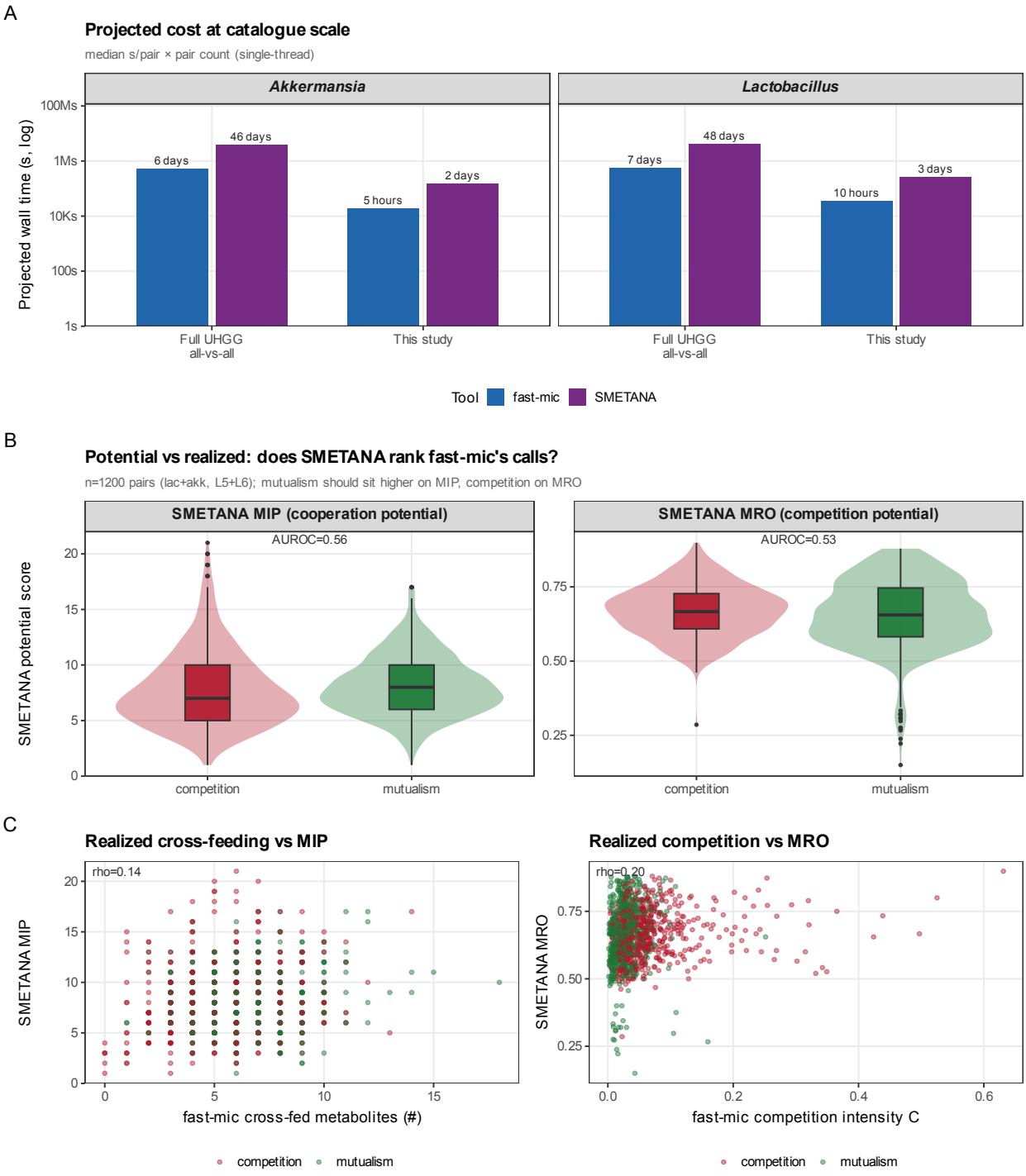

A
Projected cost at catalogue scale
median s/pair × pair count (single-thread)
Akkermansia
Lactobacillus
Projected wall time (s, log)
100Ms
1Ms
10Ks
100s
1s
6 days
46 days
5 hours
2 days
7 days
48 days
10 hours
3 days
Full UHGG all-vs-all
This study
Tool
fast-mic
SMETANA
B
Potential vs realized: does SMETANA rank fast-mic's calls?
n=1200 pairs (lac+akk, L5+L6); mutualism should sit higher on MIP, competition on MRO
SMETANA MIP (cooperation potential)
AUROC=0.56
SMETANA MRO (competition potential)
AUROC=0.53
SMETANA potential score
competition
mutualism
C
Realized cross-feeding vs MIP
rho=0.14
SMETANA MIP
fast-mic cross-fed metabolites (#)
Realized competition vs MRO
rho=0.20
SMETANA MRO
fast-mic competition intensity C
competition
mutualism